



Ateneo de Zamboanga University
College of Science, Information Technology, and Engineering
Computer Science Department



BARANGAY DISASTER SITUATION RECORD MANAGEMENT AND SITUATION REPORT GENERATION SYSTEM

A Capstone Project Proposal
submitted to the Computer Science Department
as partial fulfillment of the requirements for the degree of
Bachelor of Science in Information Technology

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INTRODUCTION

1.1 Project Context

The Zamboanga City Disaster Risk Reduction Management Office (ZCDRRMO) is the primary institutional body mandated to lead, coordinate, and institutionalize disaster risk reduction and management efforts in the city of Zamboanga. The office was established pursuant to the Republic Act No. 10121 otherwise known as “The Philippine Disaster Reduction and Management Act of 2010”, dated in May 27, 2010 and designating the members thereof as well as the City Ordinance No. 433 series 2016 (*Office of the City Disaster Risk Deduction Management | Zamboanga Website*, n.d.). This caused a shift in the country’s disaster management paradigm from purely reactive emergency response to a proactive, risk-reduction and resilience-building approach. The office operationalizes national disaster risk reduction management policies at a local scope, ensuring that the strategies are placed and contextualized to the city’s environmental and socio-economic conditions.

Guided by its vision of “Safer, adaptive and disaster resilient Zamboangueño communities towards sustainable development”, the ZCDRRMO guides the city to its vision of anticipatory and adaptive resilience. The organization’s vision reflects a commitment to safeguarding lives, protecting the environment, strengthening institutional and local capacities, and ensuring that development gains are not simply swept by disaster.

The mission of the ZCDRRMO, according to the *Office of the City Disaster Risk Deduction Management | Zamboanga Website* (n.d.), is rooted on five major pillars that are based on the four DRRM thematic areas: Disaster Prevention and Mitigation, Disaster Preparedness, Disaster Response, Disaster Rehabilitation and Recovery, and Public Safety.

In Disaster Prevention and Mitigation, the office seeks to avoid hazards and reduce their potential impacts by reducing vulnerabilities and exposure and enhancing capabilities of the communities. In Disaster Preparedness, the office aims to establish and strengthen capacities of communities to anticipate, cope, and recover from the negative impacts of the emergency occurrences and disasters. Under Disaster Response, it ensures life preservation and meets the basic subsistence needs of affected populations based on acceptable standards during or immediately after a disaster. Under Disaster Rehabilitation and Recovery, the office restores and improves facilities and living conditions and capacities of affected communities and reduces risks in accordance with the “building back better” principle. Complementing the four DRRM thematic areas the city is compliant to is Public Safety mandate, which aims to maintain peace and order, to prevent and suppress lawlessness, disorder, riot, violence, rebellion, or sedition for the generate welfare and its habitants. Crucial to this initiative is the development of a Command Center System (C2) and Incident Response Management System (IRMS) to enhance real-time monitoring, coordination, and interagency response capabilities.

To effectively deliver its mandate, the ZCDRRMO is organizationally structured into three major departments: Administration and Training, Research and Planning, and Operations and Warning Staff.

The administration and training department is responsible for the provision of administrative and capacity-building services. This division organizes and conducts training, orientation, and knowledge management activities on Disaster Risk Reduction and Management and Public Safety (DRRMPS) at the local level, including the barangays. It identifies, assesses, and manages hazards, vulnerabilities, and risks within the locality

and implements cost effective risk reduction measures and strategies. The division disseminates information and raises public awareness about hazards and risk mitigation while ensuring compliance with RA 10121 and other relevant legislative provisions. It takes continuous steps to maintain and provide suitably trained and competent personnel for effective civil defense and DRRMPS implement in the city. Furthermore, it organizes, trains, equips, and supervises local emergency response teams and Accredited Community Disaster Volunteers (ACDVs), ensuring that humanitarian aid workers are prepared for deployment. Special emphasis is placed on training barangays and the most vulnerable sectors such as women, children, senior citizens, and persons with disabilities (PWDs), particularly in disaster preparedness and risk reduction initiatives. It also implements policies and approves plans and programs of the Local Disaster Risk Reduction and Management Council (LDRRMC) consistent with national guidelines.

The Research and Planning Division serves as the strategic and policy development arm of the office. It designs programs and coordinates DRRM activities consistent with national standards and guidelines. The division consolidates local disaster risk information, including data on natural hazards, vulnerabilities, and climate change risks, and maintains an updated local risk map. It conducts research and development initiatives on DRRM and Climate Change Adaptation (CCA), while maintaining comprehensive databases of human resources, equipment inventories, directories, and the locations of critical infrastructures such as hospitals and evacuation centers. Acting as the Secretariat and executive arm of the LDRRMC, the division recommends the enactment of local ordinances consistent with RA 10121 and facilitates risk assessments and contingency planning at the local level. It formulates and implements a comprehensive and integrated local DRRM plan aligned with

national, regional, and provincial frameworks in close coordination with Local Development Council (LDC). Additionally, it prepares and submits the annual LDRRMPSO Plan and Budget, including the programming of the Local Disaster Risk Reduction and Management Fund (LDRRMF), and ensures transparency by submitting utilization reports to the Commission on Audit (COA), the Office of Civil Defense (OCD), and the Department of the Interior and Local Government (DILG). The division actively involves barangays and vulnerable sectors in risk assessment, vulnerability analysis, and planning processes to promote inclusive and participatory governance.

The Operations and Warning Staff constitutes the frontline emergency response and monitoring unit of the ZCDRRMO. It operates a multi-hazard early warning system linked to disaster risk reduction mechanisms, providing accurate and timely advisories to national and local emergency response organisations and the general public through diverse communication platforms, including radio, landline systems, and community-based technologies. The division manages and responds to the adverse effects of emergencies, ensuring efficient delivery of food, shelter, and medical supplies, particularly for women and children. It endeavors to create safe and supportive spaces for internally displaced mothers, offering assistance in breastfeeding, infant care, and peer support. The division establishes and maintains DRRM Operations Centers at the provincial, city, municipal, and barangay levels and continuously monitors disaster situations. It mobilizes instrumentalities of the LGU, civil society organizations (CSOs), private groups, and organized volunteers to utilize their facilities and resources for life and property protection during emergencies. Furthermore, it strengthens partnerships and networks with other LGUs, private sector entities, and volunteer organizations to enhance coordinated disaster

response and recovery efforts. Special attention is consistently given to the needs of vulnerable sectors, ensuring inclusive and equitable emergency management.

ZCDRRMO DEPARTMENTS		
Department	Description	Tasks
Administration and Training	Responsible for the administrative and capacity-building services, focusing on DRRMPS training, risk assessment, and personnel competency at the local and barangay levels	1. Conduct DRRMPS training and orientations 2. Identify and manage local hazards and risks 3. Public awareness and information dissemination 4. Maintain trained personnel and civil defense staff 5. Supervise emergency response teams and ACDVs 6. Focus training on vulnerable sectors 7. Implement LDRRMC policies and programs
Research and planning	Serves as the strategic and policy development arm, managing disaster risk information, local ordinances, and the comprehensive local DRRM plan	1. Design and coordinate DRRM programs 2. Consolidate risk data and maintain local risk maps 3. Conduct DRRM and Climate Change research 4. Maintain database of resources and infrastructure 5. Formulate LDRRM plans and budgets 6. Ensure transparency and COA/DILG reporting 7. Facilitate inclusive risk assessment and planning

Operations and Warning	Serves as the frontline emergency response and monitoring unit, operating early warning systems and managing disaster-time logistics and mobilization	1. Operate multi-hazard early warning systems 2. Provide timely public advisories and warnings 3. Manage emergency logistics 4. Support internally displaced families 5. Maintain DRRM operations centers 6. Mobilize LGUs, CSOs, and volunteers for response 7. Strengthen inter-LGU and private sector networks 8. Ensure inclusive emergency management
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Figure 1.1.1 ZCDRRMO Departments Profile

A Situation Report is a government, Nation Disaster Risk Reduction Management Office mandated report that is to be done by the different DRRMC offices across all the regions of the country when a disaster occurs. Private contents of the Situation Report is kept confidential to protect the privacy of the affected population while rudimentary information and data are publicized.

Current Process of ZCDRRMO Reporting and SitRep Generation

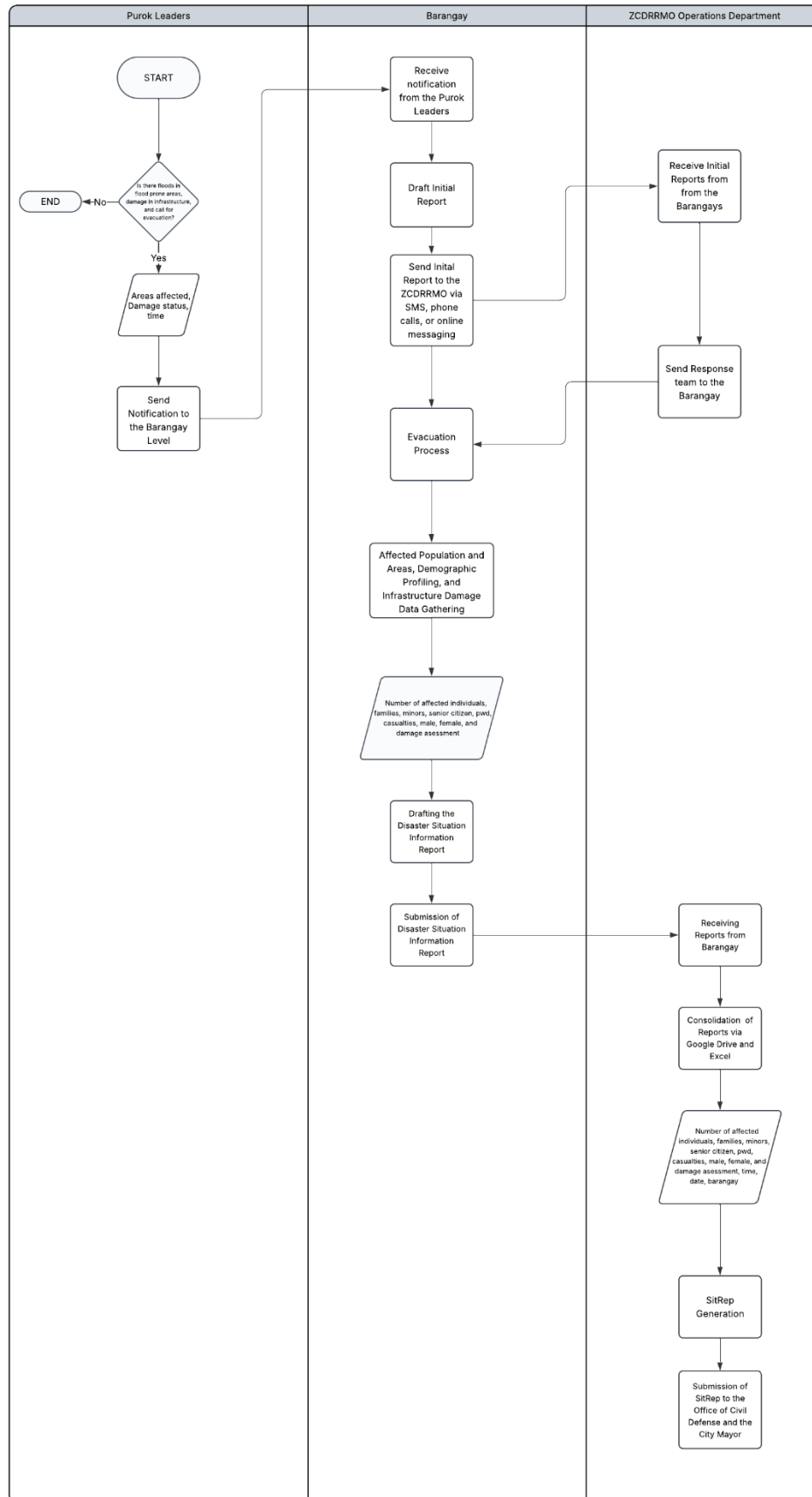


Figure 1.1.2 Current Process of Barangay Reporting and ZCDRRMO Situation

Report Generation

Figure 1.1.2 shows the current process for reporting of disaster incident data and the generation of situation reports. The process begins when the purok leaders of the barangay notify the barangay focal person for disaster, this is dependent on each barangay whether it is their Barangay Disaster Risk Reduction Management Chairman (BDRRMC) or the Barangay Captain. There are several possible bases on such a report such as floodings in flood prone areas, damage in infrastructures, and a need for evacuation and rescue. When the Barangay's focal person is notified, mobilization of rescue and on-site respondents occurs on the barangay level. The total mobilization of the personnels from the Operations Department of the ZCDRRMO should happen when multiple barangays are involved, however that is not the case today as they are also helping regardless of the number of barangays affected. Also during this time of response, drafting the initial report to be sent to the Operations Department of the ZCDRRMO is done by the personnels in each respective barangays. This report contains initial information such as affected areas, the initial damage assessment, and the time of report. The data gathering for this report is done on the evacuation centers for each respective barangays. It was also mentioned that the barangays have This report is what is sent to the ZCDRRMO via text message, online messages through messenger, or a case-to-case basis data sheet dependent on the Barangay. The consolidation of the reports from different barangays, time, and events all happen in the Operations Department of the ZCDRRMO. They are utilizing excel spreadsheets and Google Drive as their main storage for Situation Report Generation and

Data Storage. After consolidating the data, the next step for the department is the generation of the Situation Report. The completed report will then be forwarded to the Office of the City Mayor and the Office of The Civil Defense.

1.2 Statement of the Problem and Purpose

The efficacy of the office's disaster response is fundamentally tied on the velocity, integrity, precision and accuracy of disaster incident data. Currently, the office operates with a fragmented reporting system that suffers from several vulnerabilities. The core challenge in the current infrastructure is the lack of standardization in the linear progression from incident notification to formal documentation. While the initial reporting from the Purok Leaders to the Focal Person of the Barangay is functional as well as notifying the ZCDRRMO, the subsequent drafting, consolidation, and generation of the mandated Situation Report remains fragmented.

Currently, the process begins with an informal notification from the Barangays sent via SMS or online message to signal that help is needed in the barangay. This process is no doubt effective for immediate awareness, this initial report phase lacks standardization and data structure. Essential details in the initial reports are often transmitted in fragmented messages. This process forces the ZCDRRMO Operations Department to engage in back-and-forth communication to extract the specific data points required for a formal Situation Report. This delay in information consolidation is crucial during the hours of disaster.

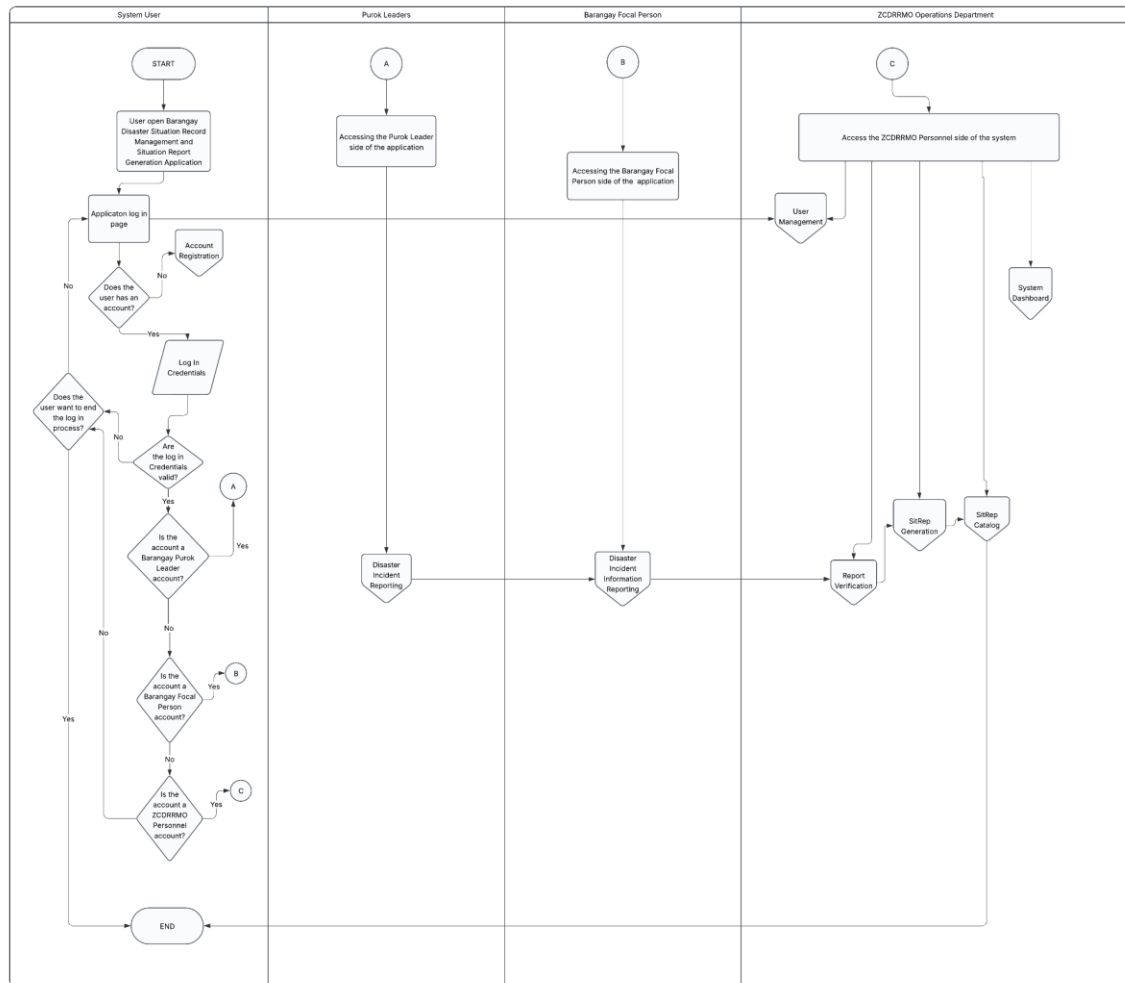


Figure 1.2.1 Proposed Online Disaster Situation Reports Generation and Record Management Flowchart

Figure 1.2.1 presents the proposed Barangay Disaster Situation Record Management and Situation Report Generation System Application Cross-Functional Flowchart, which delineates the systematic interactions between the Purok Leader, Barangay Focal Person, and the ZCDRRMO Operations Department. The flowchart illustrates the sequential logic of the system access, role-based authentication, and the functional boundaries of the application's core modules.

The entire process begins when a System User initializes the application. Upon entry, the user is directed to the Application Log In Page. When the user continues to log in, the system verifies if the user already has an account. Users without an account are directed to the Account Registration Process. Here, the user can register as a Purok Leader, Barangay Focal Person, or as a personnel of ZCDRRMO. After registration, an account verification will be initiated by the ZCDRRMO Personnels in the system. Without verification, proceeding to log in is prohibited. Verification will be placed in the system as well as the user's entered email address. Upon confirmation, the user returns to the log in page to enter their credentials. The system validates the credentials. If the credentials are incorrect, the user may retry or exit the application.

Upon the validation of user credentials, the system identifies which side of the system the user will be redirected to. If the account is for Purok Leaders and for Barangays, the user will be redirected to their respective interfaces via connector A and B. For the Purok Leaders interface, they can only access their respective reporting interface. Their reports will be forwarded to their respective Barangay Focal Person. This is marked as the initial reporting. For the Barangay Focal Person side, they can access the initial reports and notifications from the Purok Leaders. They can forward these reports to the ZCDRRMO to ask for emergency response as well as for the data gathering.

If the account is identified as a ZCDRRMO Personnel, the user is redirected to the ZCDRRMO-side through the connector C. The user will first be presented with the system dashboard that has the overview of the system details. Account Management is also handled in this side of the system which manages the user accounts. The ZCDRRMO side also has the capability to view initial reports from the barangays and incident reports from

the Barangays. These modules allow them to respond to disaster situations and to generate reports for each disaster situation. The internal connectors demonstrate a linear data progression from the Purok Leader to the City Office. This ensures that disaster records are consistent and verifiable across all levels. Upon operations are complete, the session is then ended.

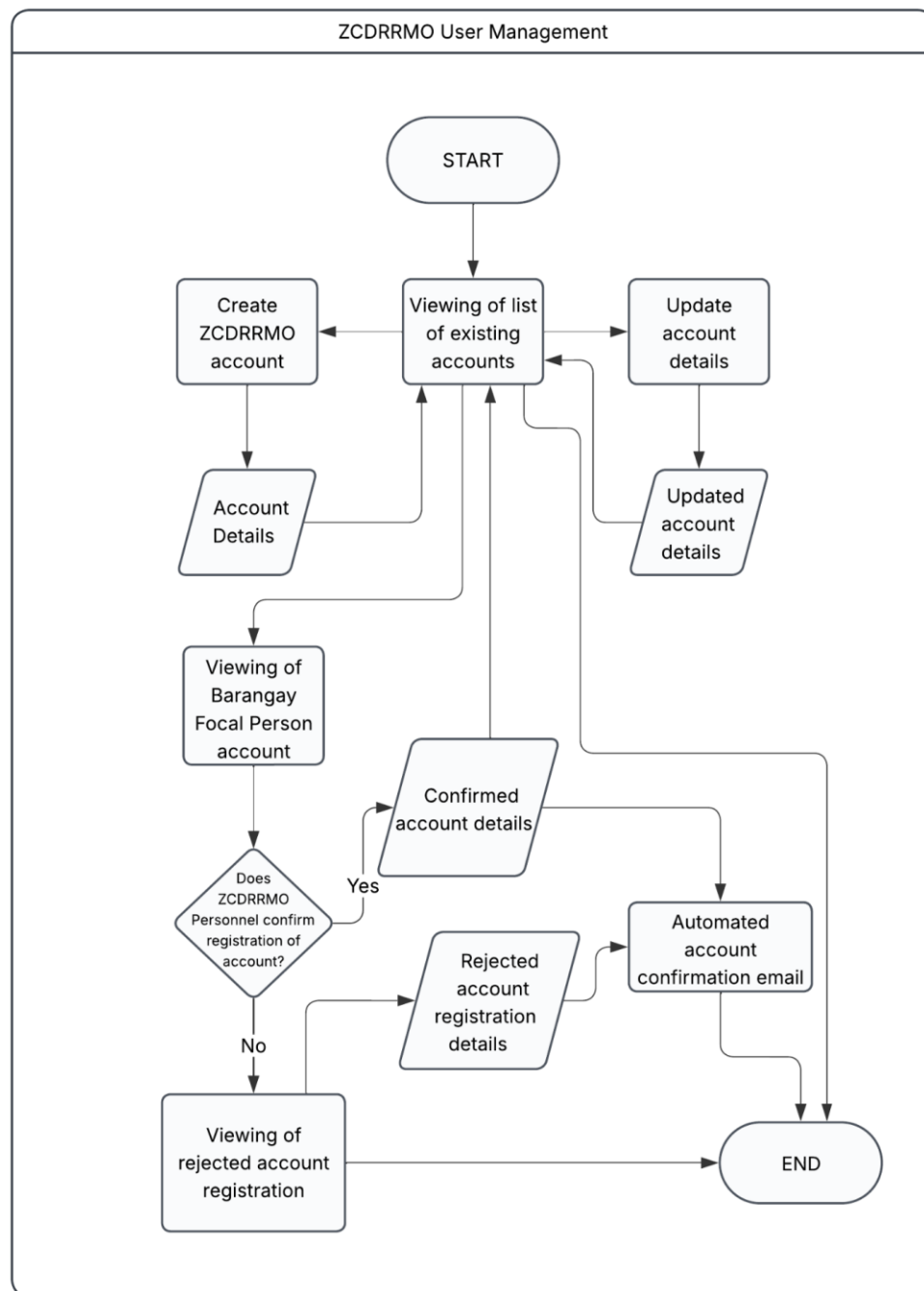


Figure 1.2.2 Proposed User Management Module Flowchart

Figure 1.2.2 shows the user management module for the proposed system. The process starts with viewing of the list of existing accounts, followed by the process of creating a ZCDRRMO account if need be, updating account details, or viewing of barangay focal person accounts. The creation of a new account will require the account details including credentials such as name and designation/position. The same goes for updating account details, it will require updated account details.

For the process of viewing a barangay focal person account, it will include the verification and confirmation from a ZCDRRMO personnel if the account and its details are proven to be legitimate, wherein if it was, the ZCDRRMO personnel will confirm the account registration and an automated account confirmation email will be sent to that user. Otherwise, the ZCDRRMO personnel would have to reject the account registration and again automatically send an account rejection (confirmation) email. Rejected account registration can also be viewed by the ZCDRRMO personnel.

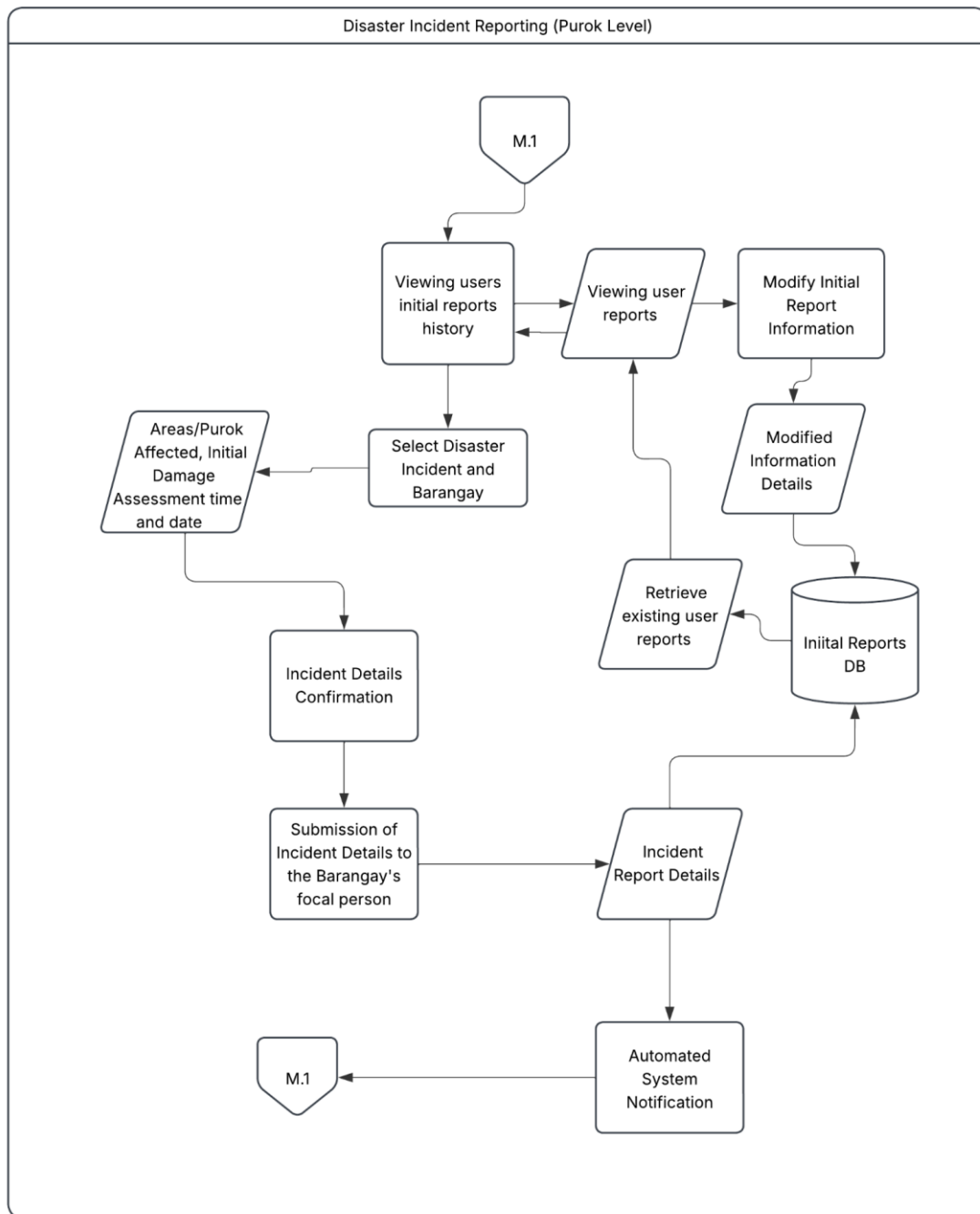


Figure 1.2.3.A Purok Level Level Disaster Reporting Module Flowchart

Figure 1.2.3.A shows how the reporting module is facilitated for the Purok Leaders. This is the initial data gateway for the entire process. The reporting module first lets the user show the reports made by the user and their status. If the user chooses to create a new initial report, the user is then prompted to enter disaster types such as floods, fire, earthquake, etc. Inclusion of affected areas, purok name, initial damage assessment, the time and date of reporting is also mandatory. After providing information for the mentioned fields and confirmation of initial report information, the initial report will then be forwarded to the Barangay Focal Person. If the user chooses to edit a report, the user will be prompted to enter change to the reports and this will be forwarded to the Barangay Focal Person.

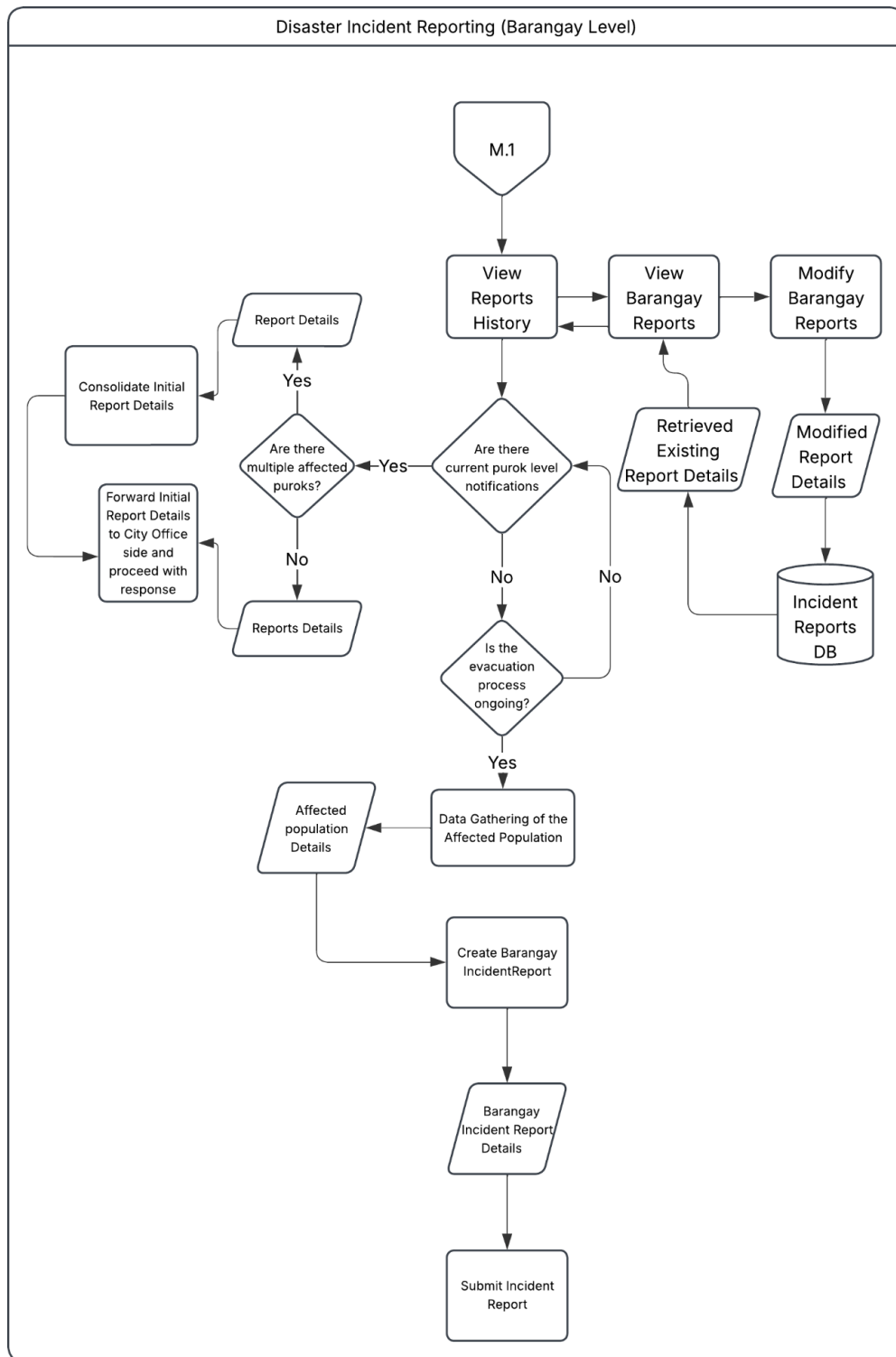


Figure 1.2.3.B Proposed Barangay Level Disaster Reporting Module Flowchart

Figure 1.2.3.B illustrates the logic of the Disaster Reporting module for the Barangay Level. This module is to ensure that the initial data from the Puroks are systematically sorted, validated, consolidated and brought up to the ZCDRRMO appropriately

The entire process begins at the initialization of the interface which starts with viewing the reports history made by the Barangay focal person. Upon entry, the system retrieves and displays existing records for the Incident Reports and Initial Reports. This also allows authorized personnel to view or modify Barangay Reports if need be, ensuring that centralized database reflect the most current occurrences. Any modifications made. Modifications are processed into the database with marks that it has been modified to maintain an accurate audit trail of the disaster's progression.

The module's logic is a series of conditional checkpoints designed to categorize the scale of the disaster incident. The system checks whether there are any Purok-level notifications. If such notifications exist, the system then assesses the scope by determining if multiple Puroks are involved. If multiple Puroks are involved, the system consolidates the report details to provide a unified situational overview. On the other hand, if the incident is isolated, the Report Details are processed individually. In both scenarios given, the system automates the forwarding of Initial Report Details to the City Office, reducing the latency between local Barangay-level detection and city-wide emergency response.

The module also facilitates another reporting function done during disaster recovery. When evacuation is ongoing, the workflow then transitions to a Data Gathering phase. Field workers input affected population details, which are then

synthesized into the Barangay Incident Report. This document undergoes a final validation before the submission of it is done. This report is also committed to the incident report database for long-term monitoring and disaster analysis. The process cycle concludes by returning to the M.1 connector which accounts for continuous updates as the disaster landscape shifts.

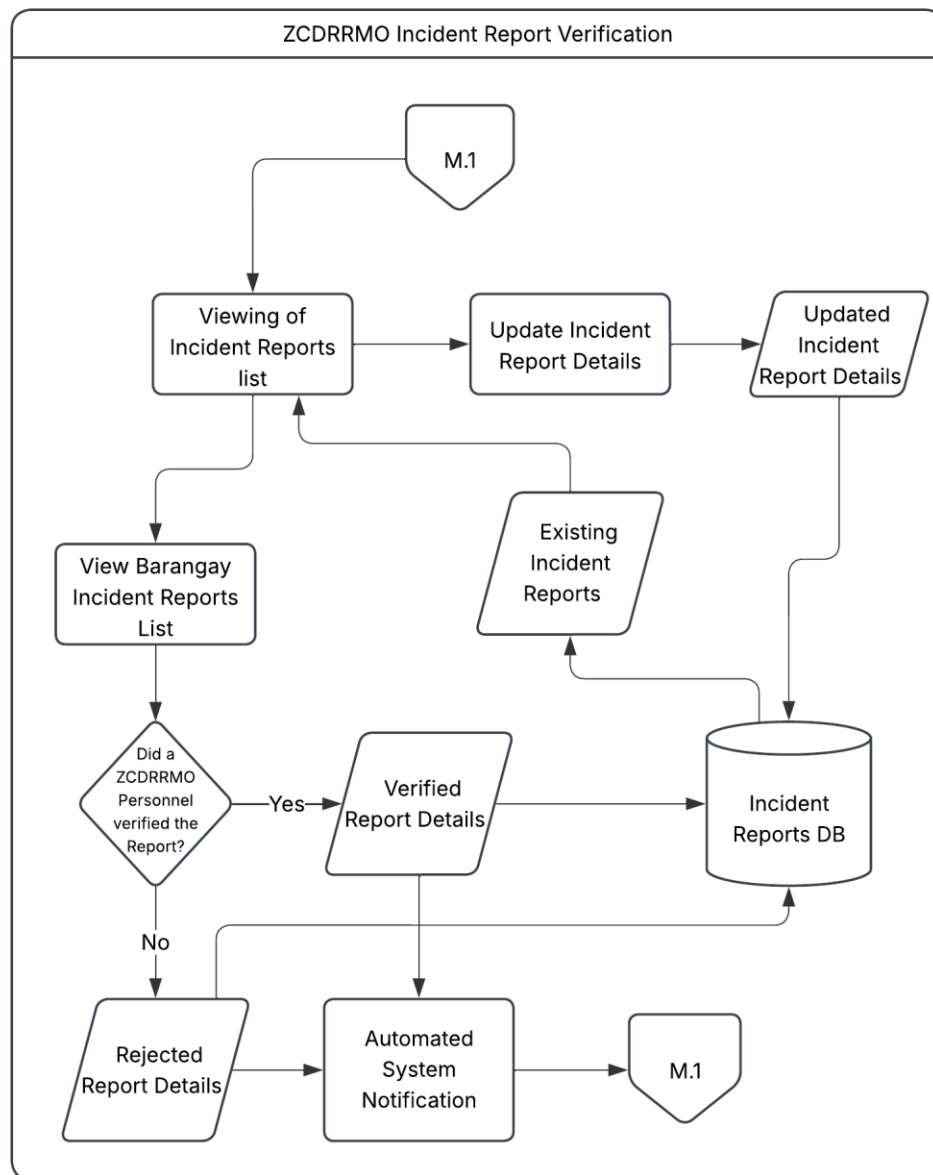


Figure 1.2.4 Proposed Disaster Report Verification Module Flowchart

Figure 1.2.4 presents the disaster report verification module. The process starts with the viewing of the incident reports list where the ZCDRRMO ground personnel-assigned and other authorized ZCDRRMO personnels to verify the reports can view the list of Barangay Incident Reports and update the incident report details if necessary. When the list is viewed, a check will be done whether a specific report has been verified by a ZCDRRMO personnel. If a report has been verified, the verified report details will be stored in the Incident Reports database, and at the same time, do an automated system notification regarding the system action. Otherwise, the report details may be rejected and again, an automated system notification will notify the rejection. Also, the existing incident reports will also be displayed during the viewing of the incident reports list.

On the other hand, the process of updating the incident report details allows the authorized ZCDRRMO personnels to edit and update the incident reports. The updated version will then be stored in the Incident Reports database. The existing incident reports will then reflect upon viewing it.

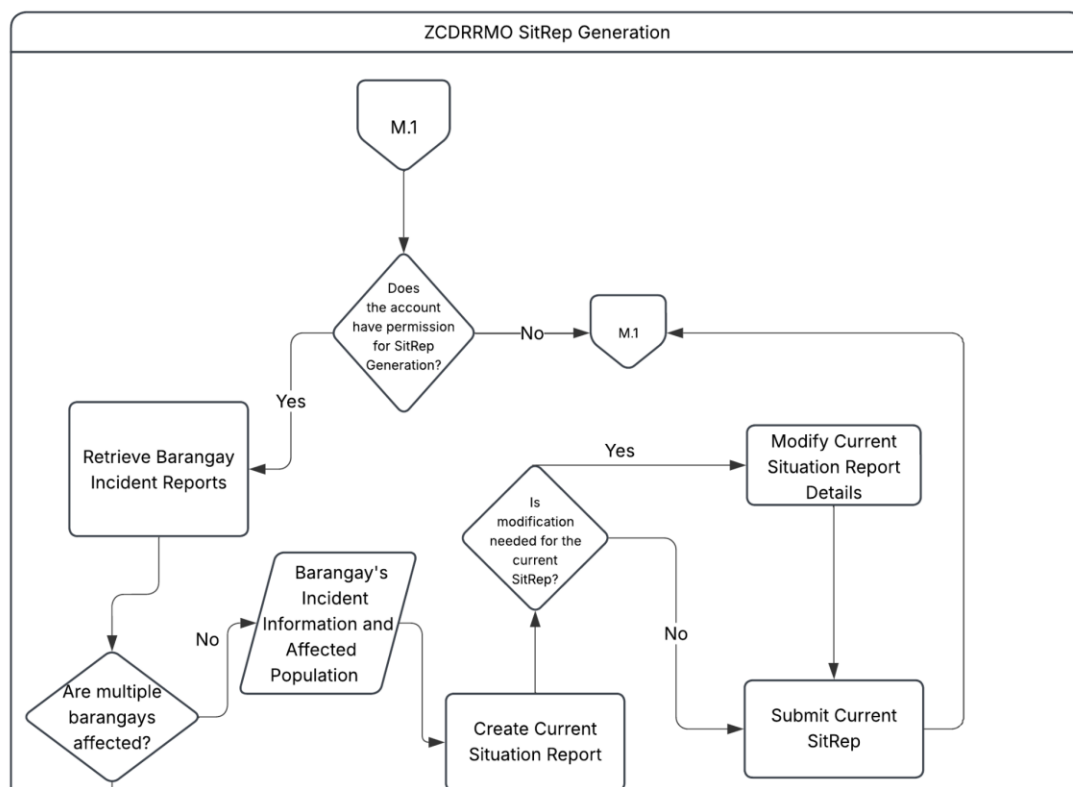


Figure 1.2.5 Proposed SitRep Generation Module Flowchart

Figure 1.2.5 shows the SitRep Generation module. The process goes by first checking whether the account has permission to generate a SitRep. If the account is not allowed to generate SitReps, then the user will not be able to access the generation of SitReps. Otherwise, the process starts with retrieving barangay incident reports. If there are multiple barangays affected by the disaster, the information from the incident reports will be consolidated then the current situation report will be created based from the consolidated barangays incident information, including affected population. Otherwise, if only one barangay is affected by a disaster, the process moves with the creation of the current situation report for that barangay based on the barangay's incident information and affected population data. After consolidations and creation of the situation report, if modifications have to be made, the system will allow the authorized ZCDRRMO personnels to modify the current SitRep. Otherwise, the SitRep will be submitted to the

City Mayor's Office for immediate public awareness and the Office of the Civil Defense for the publication of the SitRep for the general public later on.

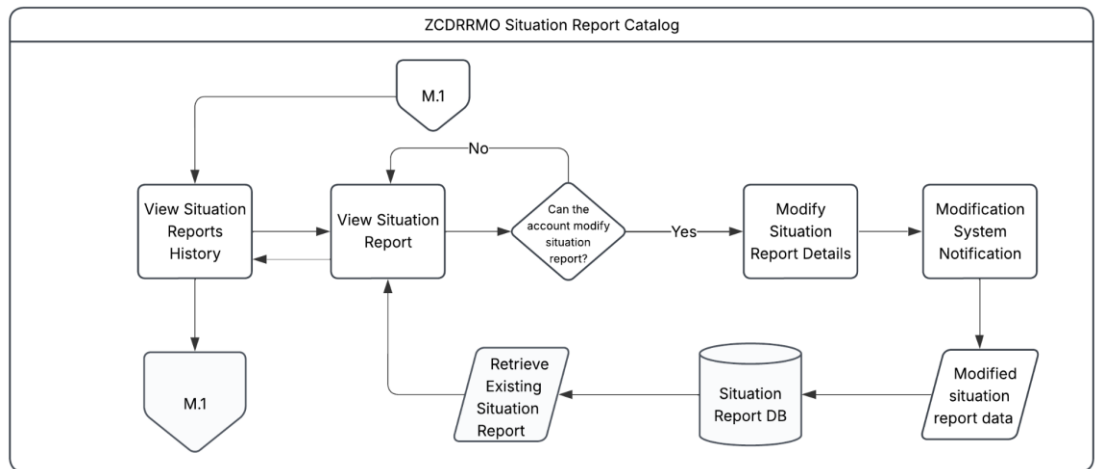


Figure 1.2.6 Proposed SitRep Catalog Module Flowchart

Figure 1.2.6 shows the SitRep Catalog Module Flowchart of the proposed system. This flowchart describes how authorized ZCDRMMO personnels can view and edit the generated Situation Reports within the system. Specifically, the situation reports history can be viewed and from that, a specific situation report can be viewed by the users authorized to access the situation reports. If the account can modify or edit the situation report, they can proceed to modify the necessary details that need modification (e.g., corrections), otherwise, they are only allowed to view the SitReps. For each modification, the system will notify the user that the report has been modified and that the edited version will be saved in the Situation Report database.

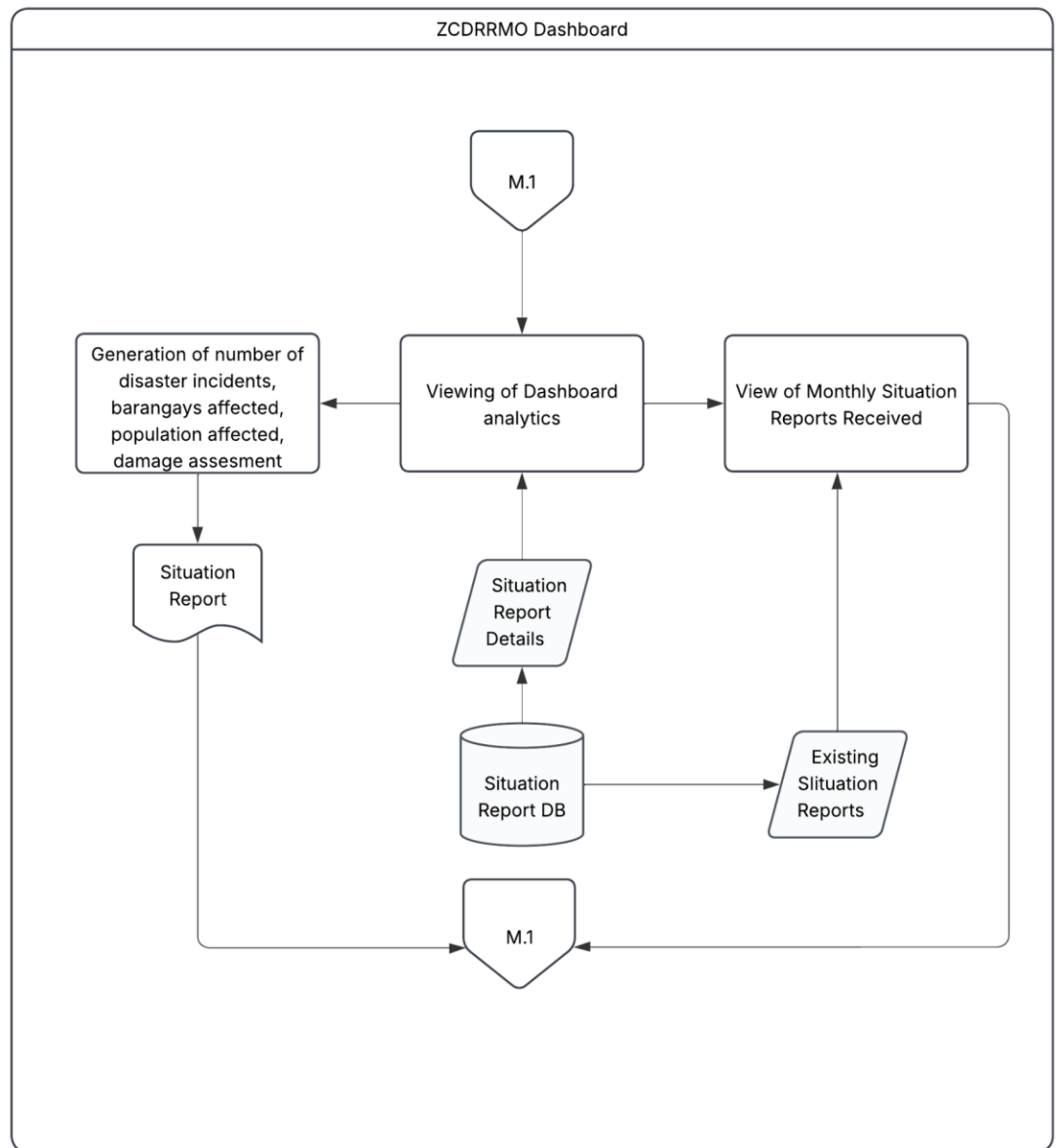


Figure 1.2.7 Proposed Dashboard Module Flowchart

Figure 1.2.7 shows the processes in the Dashboard Module of the system. This process will allow the authorized users, especially the ZCDRRMO personnels, to view relevant and up-to-date analytics regarding the number of disaster incidents, barangays

affected by the disaster, population affected, and the damage assessment. It serves as the primary analytical command center for authorized personnel.

The process initiates via the M.1 connector, which transitions the user to viewing the dashboard analytics interface. Upon initialization, the system interfaces with the Situation Report DB to fetch current Situation Report Details. These data are processed to generate real-time metrics, including the total number of disaster incidents, the specific barangays impacted, cumulative and dissected affected populations, and comprehensive damage assessments. This centralized view ensures that decision making can be done with an immediate access to critical information regarding the city's current landscape in a disaster situation.

In addition to real-time analytics, the module supports the Generation of Situation Reports. By aggregating the aforementioned metrics, the system compiles formal documentation that encapsulates the scope of the disaster, providing a structured output for administrative view. Furthermore, the dashboard also has the capability to view monthly situation reports received and submitted. By retrieving existing situation reports from the database, personnel can perform comparisons and response efficiency over a selected period of time. After the aforementioned functions, the process returns to the M.1 connector which allows for the continuous monitoring of incoming data streams.

1.3 Objectives

General Objective

This research project aims to develop a Barangay Disaster Situation Reports Record Management and Generation System for the Zamboanga City Disaster Risk Reduction Management Office.

Specific Objectives

1. To develop a *User Management* for the ZCDRRMO to ensure sensitive information like Personal Identifiable Information (PII, e.g., Full Name) are kept confidential from unauthorized access, ensure and establish authority to information hierarchy, and to limit access only to authorized personnel like Purok Leaders or equivalent representatives, Barangay Officials (e.g., Barangay Captain, Barangay DRRMC Chair), and ZCDRRMO personnel.
2. To develop a *Disaster Reporting Module* that will allow Purok Leaders to notify the Barangay Captain or BDRRMC Chair of an alarming disaster like high flood levels that needs immediate attention. Also, to allow the Barangay Captains/BDRRMC Chair to submit an initial report to the ZCDRRMO Operations Department.
3. To develop a *Disaster Report Verification Module* that will allow for organized verification of disaster incident reports by the ZCDRRMO personnels.
4. To develop a *SitRep Generation Module* where reports are constantly updated automatically based on the entry from the barangay/s and upon

verification and confirmation of the ZCDRRMO verification personnel-assigned.

5. To develop a *SitRep Catalog Module* that will store and chronologically categorize the Situation Reports according to disasters. This allows the related ZCDRRMO departments like the Research and Planning to quickly access and search for a specific report.
6. To develop a *Dashboard Module* that will display relevant summaries regarding SitRep and allows ZCDRRMO personnels to read and filter analytics of Disaster Situation Reports.

1.4 Scope and Limitations

The Barangay Disaster Situation Reports Record Management and Generation System for the Zamboanga City Disaster Risk Reduction Management Office is an online mobile-based application accessible to purok leaders, barangay captain/BDRRMC, and ZCDRRMO personnels. The system will include the following features: *User Management, Disaster Reporting Module, Disaster Report Verification Module, SitRep Generation Module, SitRep Catalog Module, and Dashboard Module.*

The *User Management* will feature user authentication such as registration and login set-ups to ensure that the users of the system are those who are authorized to do the reporting, access confidential information like names of the victims, and generate the SitRep. This module will also be designed according to the user access hierarchy. For instance, purok leaders may only report an ongoing disaster incident to the BDRRMC chair, but cannot generate the SitRep itself.

The *Disaster Reporting Module* will serve as the primary means for the purok leaders to send an ongoing disaster incident to the BDRRMC to notify the Barangay officials (purok-level disaster incident reporting), and for the Barangay Captain or BDRRMC Chair to submit the initial and continuous reports to the ZCDRRMO (barangay-level disaster incident reporting).

On the other hand, the *Disaster Report Verification Module* will serve as a means for the ZCDRRMO to verify the initial and continuous reports submitted by the barangay captain. This will be designed to work hand-in-hand with the ZCDRRMO personnels as they do the verification. Reports may be edited and rejected by the authorized ZCDRRMO personnels. Reports will be edited in the event a report lacks other crucial information or there is additional vital information that has to be included. Also, reports will be rejected in the event it was proven to be illegitimate.

Moreover, the system will include the *SitRep Generation Module* where disaster situation reports can be reviewed and generated by the authorized ZCDRRMO personnels. This will feature automated consolidation and update of the situation report based on the submission of the barangays.

Also, the *SiteRep Catalog Module* will be included in the system where the final generated disaster situation reports will be stored in chronological order and based on the disaster category (i.e., Typhoon, Flood, and Fire). This module will feature filters and search bars for easier and optimal access. Also, authorized ZCDRRMO personnels may edit the generated Situation Reports in the event that it must be edited for correction. SitReps can also be downloaded from the system to the authorized user's or the office's local device (i.e., computer, laptop).

Lastly, the system will also include the *Dashboard Module* where the authorized ZCDRRMO personnels can view the analytics of the website including, but not limited to, initial reports entries, number of disaster incidents reported, and number of situation reports generated. This will serve as an analytical tool for the ZCDRRMO to monitor relevant and key trends using the data.

As for the limitations of this project, the system will only cover the floods, typhoons, and fire disasters and will not cover other disasters like the earthquake, landslide, tsunami, and volcanic eruption. Also, real-time operations in the system, such as disaster reporting, will mainly take place online. Factors such as network signal and internet connectivity are beyond the control of this project. Also, ZCDRRMO personnels will be responsible for checking and validating if a user (newly registered account) is legitimate and has permission to access the system, as well as for overseeing the transactions within the system (e.g., initial reporting, report verification, etc.). Ultimately, the system will be designed exclusively for the ZCDRRMO Situation Report Generation and Record Management process and will not handle the announcements to the general public.

1.5 Significance of the Project

The development of the Barangay Disaster Situation Reports Record Management and Generation System for the Zamboanga City Disaster Risk Reduction Management Office serves as a critical technological intervention in the localized disaster response framework of the city of Zamboanga. The significance of the study heavily impacted and manifested in different sectors and dimensions.

First, the system smoothenes the process of disaster response. By transitioning from manual, verbal, or unverified social media message reporting to a structured Disaster Reporting Module, the office is now able to gain real-time, high-fidelity overview of the emergencies and plan their respective responses. The Geotagging and Report Verification features mitigate the risk of inaccurate and less precise results. This helps in ensuring that the assets and resources put for such emergencies are deployed precisely. All of the mentioned benefits would be incredibly helpful for the **ZCDRRMO Operations and Warning division**.

Next would be the benefits that would be felt in the **Barangay Level**. The system provides a formal platform for accountability and transparency. It professionalizes the flow of data from the Barangay Level to the ZCDRRMO. It ensures that the localized threats and incidents are elevated to the awareness of the city-level timely by enforcing a streamlined process as opposed to the current manual process.

The system will also benefit the **Research and Planning division of the ZCDRRMO**. It transforms raw emergency data into actual information that could be utilized for intelligence. This allows for evidence-based decision making and policy making as well as hazard mapping. By maintaining a searchable database of situation reports

Moreover, the project as a whole will be beneficial to the **future researchers** in their research endeavors. The project may provide key insights in their field of study and may use this as a reference in the future to help them enrich their study and in cases of reviewing related literature. Furthermore, this research project may spark a drive in

technological advancements as technologies continue to evolve and the need for cutting-edge technologies continue to increase in the modern times.